

Kaixiang Yao

Robotics Engineering Undergraduate

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kaixiangyao.github.io

EDUCATION



Southern University of Science and Technology

Shenzhen, China · Aug. 2023-Jun. 2027

B.Eng. in Robotic Engineering · GPA: 3.91/4.0 · Rank: 1/90

Selected coursework: Fundamentals of Control Engineering (100), Mechanics of Materials (100), Probability & Mathematical Statistics (99), Theory of Machines (98), Robot Operating System (97), Mechanical Design (96), Robot Modeling and Control (94).



University of California, Berkeley

Berkeley, CA · Jun.-Aug. 2026

Visiting Undergraduate Student

Coursework: The Planets (ASTRON W12); Prototyping & Fabrication (DES INV 22).

RESEARCH FOCUS

Physically grounded robot learning: using contact, deformation, force, and task outcomes to make physical interaction safer and more adaptive.

PUBLICATIONS & MANUSCRIPTS

Gaze-Driven Perception-Action Control of Multi-Channel FES for Efficient Grasp Restoration

Kaixiang Yao, Zeyang Yang, Jinxin Sun, Kong Hoi Cheng, Junyun Fu, Zhaokai Chen, Chengyu Lin, Chenglong Fu. Under revision at *IEEE Transactions on Medical Robotics and Bionics*, 2026.

Quasi-real-time digital twin for robotic assembly: a physics-based multi-fidelity framework

Kaixiang Yao, Hao Chi, Dingye Chen, Jiwen Zhang. *Journal of Intelligent Manufacturing*, 2026.

CARTILAGE: A Convergence-Aware Rigid-Soft Contact Runtime for Batched Robot Learning

Kaixiang Yao, Chao Huang, Yanlin Liu, Yao Shen. Under review at *Conference on Robot Learning (CoRL)*, 2026.

RESEARCH EXPERIENCE



University of California, Berkeley

Berkeley, CA · Jun. 2026-Present

Visiting Undergraduate Researcher · Advisor: Prof. Masayoshi Tomizuka · Mentored by Yuxin Chen

Embodied AI for Physical Manipulation

- Developing simulation-based evaluation workflows for learning and control in contact-rich manipulation.
- Studying how physical interaction feedback can support robust behavior under changing dynamics.
- Designing controlled evaluations to identify failure modes and improve experimental rigor.



Shanghai Jiao Tong University

Shanghai, China · Oct. 2025-Present

Undergraduate Researcher · Advisor: Prof. Yao Shen

Robot Learning for High-Fidelity Rigid-Soft Contact

- Developed a simulation-and-learning framework for a four-finger PneuNet soft gripper and fragile-object grasping.
- Designed soft-gripper assemblies, pneumatic actuation semantics, contact/tactile definitions, and RL-compatible tasks.
- Studied how contact fidelity and force-safety rewards affect damage-free grasping.

Kaixiang Yao

Research Experience, Skills & Recognition

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RESEARCH EXPERIENCE



Southern University of Science and Technology

Shenzhen, China · Jul. 2024-May 2026

Undergraduate Researcher · Advisor: Prof. Chenglong Fu

| Eye-Tracking and Electrical Stimulation-Assisted Grasping

- Proposed normalized gaze-variance features and random-forest recognition with dual-threshold temporal filtering.
- Integrated YOLOv5 object detection and MediaPipe hand-keypoint estimation for object-aware grasping strategies.
- Designed an 8-channel, 54-electrode high-density FES sleeve for selective muscle activation.
- Verified approximately constant interaction time as gesture options increased, supporting O(1)-style selection behavior.



Tsinghua University

Beijing, China · Jun.-Aug. 2025

Undergraduate Researcher · Advisor: Prof. Jiwen Zhang

| Quasi-Real-Time Digital Twin for Robotic Assembly

- Built a ROS 2 non-blocking pipeline separating real-time safety control from background FEA.
- Used kinematic TIE constraints and mesh-scale analysis to improve robustness and computational predictability.
- Integrated full-field stress outputs into an FSM-based safety evaluator.

SKILLS

Programming

Python, Java, MATLAB, ROS 2

Robotics & Control

Robot modeling and control, grasp planning, safety reflex control, FSM decision logic

Learning & Perception

Reinforcement learning, YOLOv5, MediaPipe, random forests

Simulation & Modeling

MuJoCo/MJLab, FEA/FEM, rigid-soft contact, mesh-scale analysis

Design & Fabrication

CAD assemblies, PneuNet grippers, high-density FES sleeves

HONORS & AWARDS

First Prize · Provincial Level (2nd Place)	2025
Guangdong Provincial College Students Engineering Practice and Innovation Competition	
Outstanding Student Scholarship	2024-2025
Outstanding Student Scholarship	2023-2024